Félix Wilhelmy

Quantum Computing Engineer

Education

Master's Degree with Thesis 2024–2026
École de Technologie Supérieure
Quantum Machine Learning

Graduate Coursework 2025
MILA / Université de Montréal
Representation Learning (IFT6135)

AlgoLab Summer School 2024
Institut Quantique / Université de Sherbrooke
Near-Term Quantum Algorithms

Bachelor of Engineering 2018–2024
École de Technologie Supérieure
Software Engineering

DEC 2013–2016
Collège Montmorency
Computer Science

Skills

Programming Languages

Python, C++, C#, Java, SQL, TypeScript, JavaScript

Tools

PennyLane, Qiskit, PyTorch, NumPy, Hydra, Git, Docker, Linux, Bash, SSH, Pytest, Jenkins

Software Engineering

Data Structures, Algorithms, DevOps, CI/CD, Unit Testing, Software Architecture, Performance Optimization

Quantum Computing

Variational Circuits, Quantum Optimization, QAOA, VQE, QUBO, Entanglement Distillation, Photonic Circuits, Superconducting Circuits

Machine Learning

CNN, Transformers, LLMs, Generative Models, Hybrid Quantum-Classical Models, Adversarial Robustness, Model Benchmarking, Statistical Analysis

Languages

French (Native)

English (Professional Proficiency)

Contact

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About

I just completed my research master's in quantum machine learning (GPA 4.06/4.30; Excellent mention at my thesis defense). I combine quantum computing research with hands-on industrial software engineering, developing systems from scratch and solving complex technical problems. My background includes technical leadership and international quantum optimization competitions. A scientific paper based on my master's research is in preparation.

Experience

QB-PIER – Quantum & Hybrid Model Robustness 2024–2026
Master's Research Project - École de Technologie Supérieure

- Independently designed custom quantum-classical architectures using modular variational circuits, configurable ansätze, and parallel quantum processing blocks.
- Benchmarked 10 classical, quantum, and hybrid architectures across 50 trained variants through extensive PyTorch/PennyLane experiments, identifying hybrid designs with strong robustness against small adversarial perturbations.

Esports Data Analyst May 2026–Sep 2026
Canadian National Team - Overwatch World Cup 2026

- Supported Team Canada's coaching staff with performance analysis, opponent scouting, and data-driven recommendations for international competition
- Built Python/SQL analytics tools and dashboards to turn game logs into player, team, and composition metrics, comparative analyses, and recurring coaching reports

President (24–25) & VP Competitions (23–24) Sep 2023–Jan 2026
Student Club QuantumÉTS - Montréal, QC

- Led a 40-member quantum computing club, secured its first industry sponsorship, and coordinated technical workshops, competition teams, and student mentorship
- Guided teams achieving podium finishes at all six competitions attended during my presidency, including hackathons hosted by MIT, Yale, MILA, and Polytechnique Montréal

Full Stack Developer (Internship) — React/C# Sep 2022–May 2023
Amilia - Montréal, QC

- Developed and maintained customer-facing features in a production SaaS application, including gift-card functionality across C# backend services and a React frontend
- Improved software reliability and developer workflows through unit testing, Jenkins CI/CD enhancements, debugging, and onboarding documentation

Research Assistant (Internship) — C++/Networking Apr 2021–Aug 2021
LASSENA Research Laboratory - Montréal, QC

- Developed cross-platform C++/Qt research software for avionics analysis, including aircraft instrument interfaces and navigation-data integration
- Designed UDP-based communication for instrument data exchange and optimized background processing to maintain responsive interfaces under real-time constraints

Full Stack Developer (Internship) — Angular/C# May 2019–Sep 2019
Air Transat - Montréal, QC

- Accelerated a critical SQL search query by approximately 6× through Entity Framework optimization, lighter query models, and asynchronous processing
- Developed internal employee flight-request and availability tools in C#/Angular; migrated Angular 7.2 to 8 and introduced lazy-loaded modules

Awards

1st Place, iQuCodeFest 2025
Polytechnique Montréal
Quantum Computing Hackathon

2nd Place, YQuantum 2025
Yale Quantum Institute
Quantum Optimization Hackathon

Best Effort Award, Mil'HaQ Fest 2025
MILA / Université de Montréal
Quantum Machine Learning Hackathon

Projects

Adaptive Graph Strategy (Hackathon) 2026
MIT iQuHack 2026 - Cambridge, MA, USA

- Co-developed modular quantum circuits for entanglement distillation, combining X/Z parity checks to address errors in noisy Bell pairs
- Integrated adaptive circuit selection with graph-based optimization and fidelity feedback, balancing resource consumption, success probability, and network expansion

Photonic Volatility Forecasting (Hackathon) 2025
Mil'HaQ Fest 2025 - Mila / Université de Montréal

- Engineered a photonic quantum LSTM pipeline for financial volatility forecasting, using local time-series predictions compatible with quantum circuit constraints
- Implemented PyTorch training with quantum parameter optimization, chronological data splitting, and leakage-free normalization for reproducible experiments

Wildfire Resource Allocation (Hackathon) 2025
Yale YQuantum 2025 - New Haven, CT, USA

- Co-developed a quantum optimization approach for allocating limited firefighting resources across high-risk wildfire regions
- Formulated geospatial resource allocation as a constrained QUBO, integrating wildfire-risk data processing with QAOA and classical reference optimization

Additional Experience

Teaching Assistant Jan 2024–Aug 2025
École de Technologie Supérieure - Montréal, QC

- Supported quantum-computing laboratories and introductory machine-learning teaching
- Guided quantum-algorithm implementation and debugging; prepared LaTeX technical summaries and teaching material

Esports Coach 2016–2026 (Intermittent)
International Competitive Teams - Overwatch

- Coached professional and competitive esports teams internationally, leading strategic preparation, performance reviews, and player development